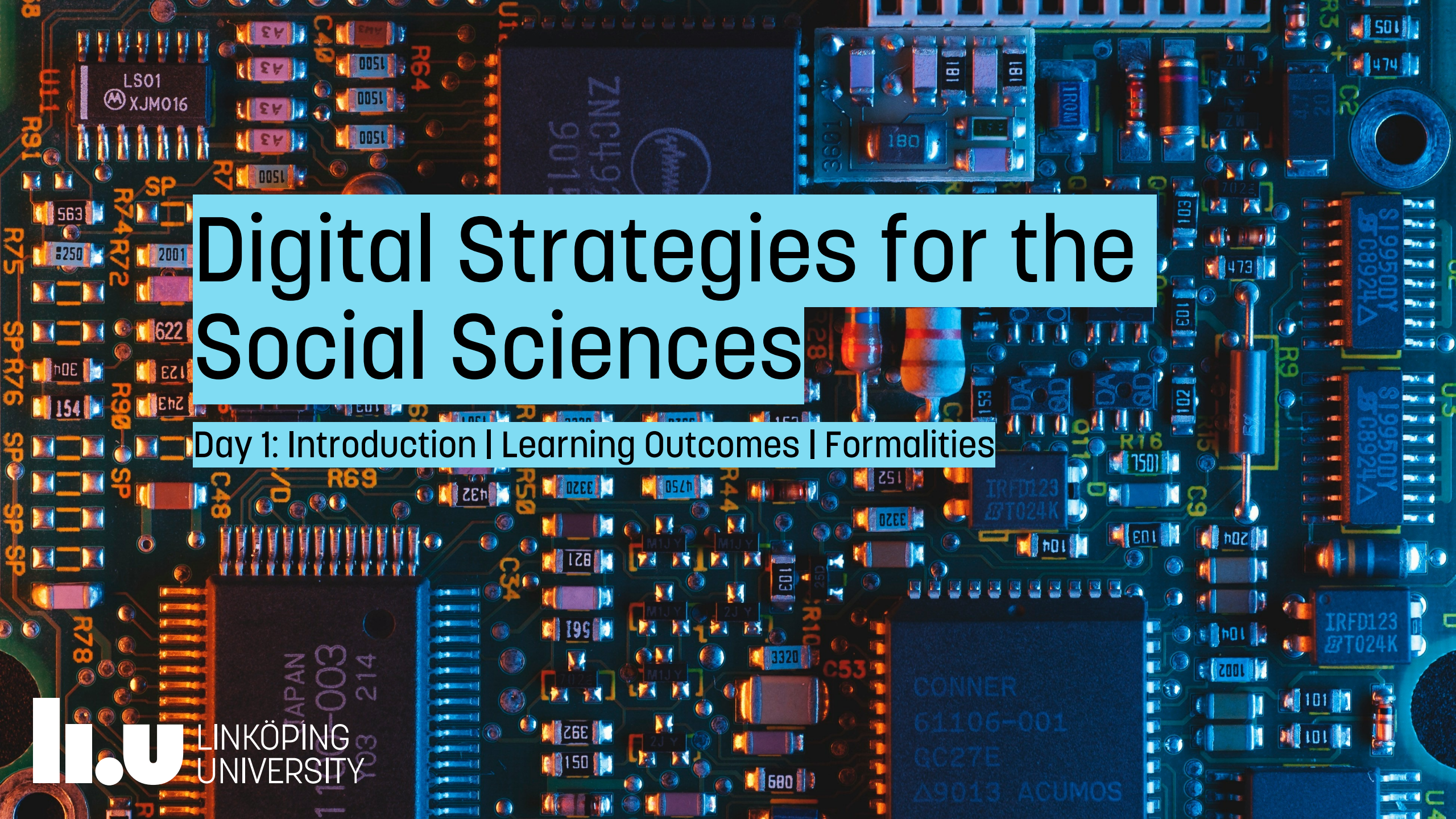




Digital Strategies for the Social Sciences

Day 1: Introduction | Learning Outcomes | Formalities

Previous Evaluation & Changes

- Last year's evaluation gave the course very high marks in all categories
- Average total score was 4.7 out of 5 (n=50 participants)
- I strive to provide the same quality this time

Previous Evaluation & Changes

- However, there has been one big change: Last year, Étienne Ollion was giving this course, this year, he gave the course to me
- What I leave in place:
 - The examination, that is, the work you will have to do, remains the same as before
 - The labs will remain as they are (with minor improvements in legibility and accessibility where necessary)

Previous Evaluation & Changes

- What I will change:
 - I have entirely different slides since slides are very personal and need to work for the instructor
 - They still contain mostly the same information, except for:
 - I include more info about APIs and Selenium, since that will be more important for the future (= provide more value to you)
 - I try to streamline the information and reorganize it a bit so that it makes more sense
 - I will provide more real-world examples so that you get a better feeling for how all of this looks in practice

Introduction

Introduction Round

- Dr. Hendrik Erz
- Postdoc at the Institute for Analytical Sociology
- Political Sociologist working with text data and parliamentary speech
- Focus on data pipelines and setups; have taught this course as a TA before (first time course director!)



Below: Carl Nordlund,
teaching second week



Introduction Round

- Please tell me...
 - ...who you are
 - ...your (academic) background
 - ...what you hope to get from this course
 - ...what really piques your interest about digital strategies



All comics in these slides from XKCD



Readymade

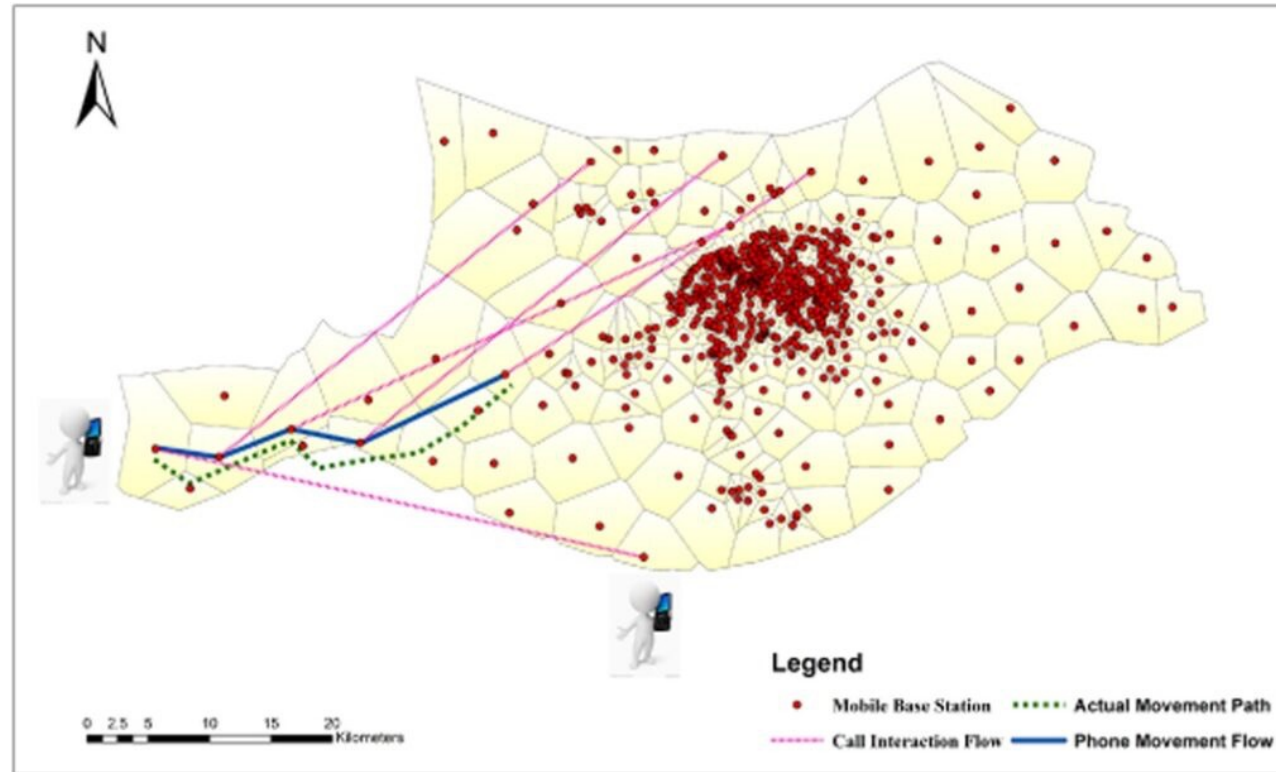


Custommade

Matt Salganik, Bit By Bit, Fig 1.2

Readymade Data Source

Human mobility
research with phone
data and geolocation.



Source: <https://geography.wisc.edu/geods/archives/582>

Custommade Data Source

Tabelle 6: Zustimmung zu Aussagen aus dem Kontext der Montagsmahnwachen (Angaben in Prozent)

Aussage	Zustimmung				
	Lehne völlig ab	Lehne überwiegend ab	Stimme teils zu/teils nicht zu	Stimme überwiegend zu	Stimme ganz zu
Amerika bzw. das amerikanische Militär ist nur der Knüppel der FED (US-Notenbank). ³⁴	1,4	7,5	39,4	30,5	21,1
Die BRD ist kein souveräner Staat. ³⁵	5,1	9,5	24,0	24,4	37,1
Die Zionisten haben sich weltweit an die Hebel der Macht gesetzt und lassen nun Politik, Börse und auch die Medien nach ihrer Pfeife tanzen. ³⁶	37,5	15,5	19,7	14,0	13,3
Friedensaktivisten werden derzeit von einer nahezu gleichgeschalteten Presse in die rechte Ecke gestellt. ³⁷	3,0	0,7	7,4	37,1	51,8

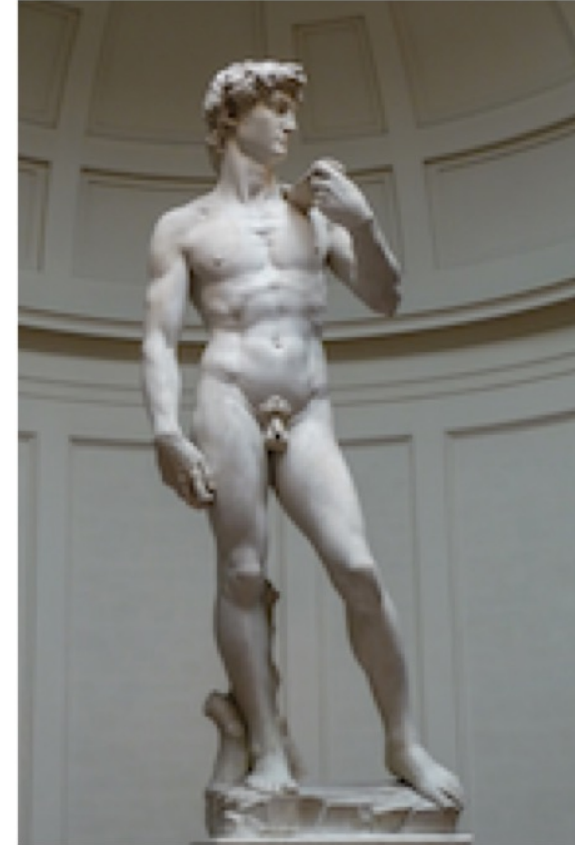
Custom survey for the
“Monday protests” in
Germany after 2014

Source: https://protestinstitut.eu/wp-content/uploads/2022/08/ipb-wp-2014.1_Occupy-Frieden.pdf

This Course



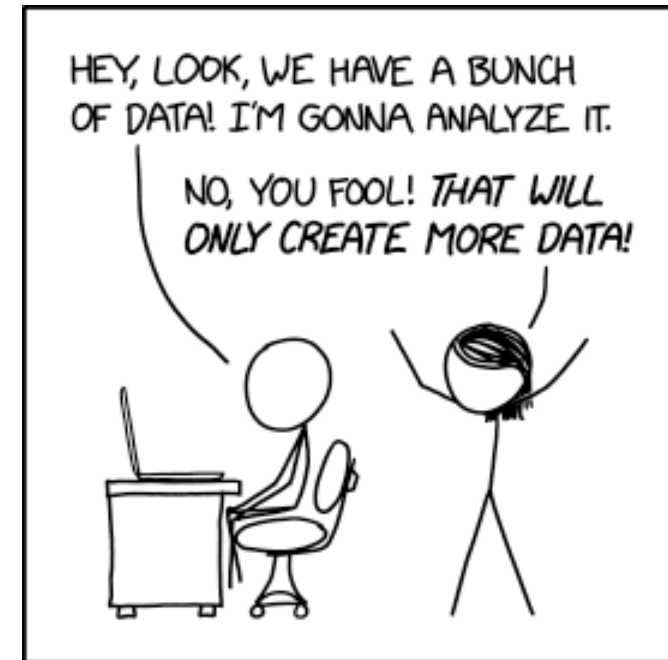
Readymade



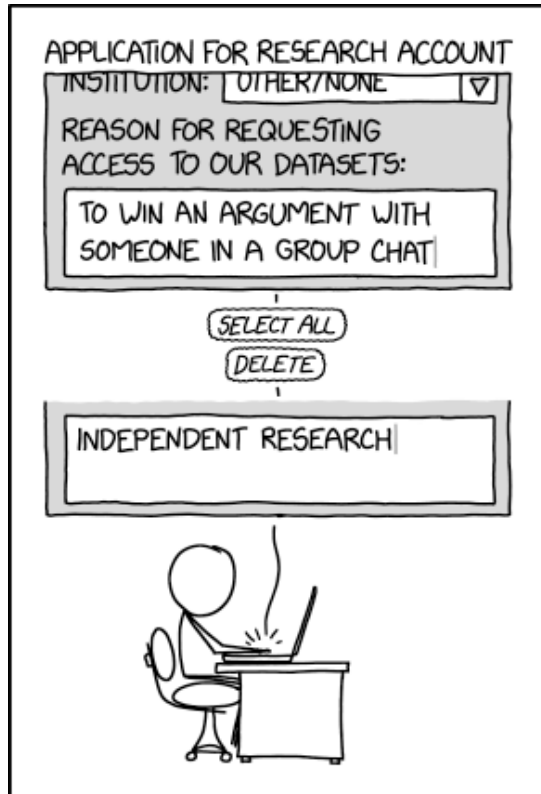
Custommade

Digital Strategies: What, Why, and How

- There are “readymade” data sources out there; existing datasets that researchers can repurpose
- There are also “custommade” data sources; datasets that researchers create from scratch
- This course is about the middle ground: existing, “readymade” data but that is not available in dataset form
- → Making “readymade” data “custom”



Digital Strategies: What, Why, and How

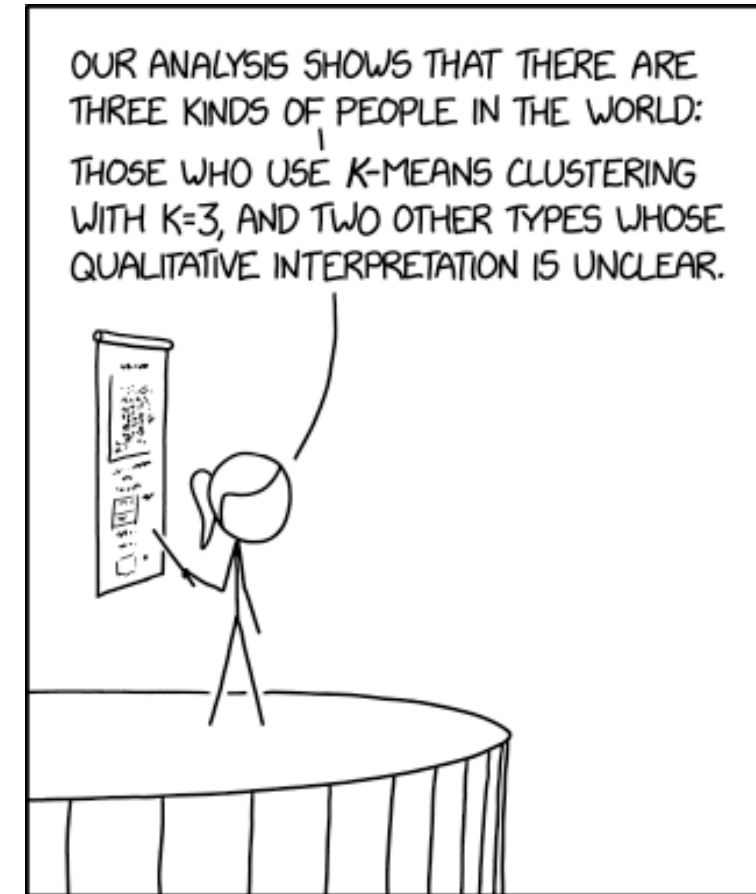


I NEVER KNOW HOW HONEST
TO BE ON THESE FORMS.

- This course will teach you how to turn data you found online into a dataset that can help you answer your research question
- It does *not* focus on how to find the data, nor how to shove it through an analysis

Digital Strategies: What, Why, and How

- The course consists of two project-weeks; intensive workshops that bring you from zero to hero in five days each
- **Week 1:** Focuses on retrieving and cleaning data
- **Week 2:** Focuses on producing descriptive statistics of newly cobbled-together datasets



Week 1 Schedule

1. Introduction
2. Getting the Data
3. Extracting the Data
4. Dealing with JavaScript
5. Regular Expressions and Cleaning



Learning Outcomes: A Case Study

How do “digital strategies” look like in practice?

Learning Outcomes: A Case Study

- A few weeks ago, I needed a dataset that contains information on **party leadership in U.S. Congress** across history
- Seems simple, right?
- Here's my journey to getting such a dataset.
- Side note: “Leadership” means:
 - Majority and minority leaders in both House and Senate
 - (Speaker of the House)
 - Democratic and Republican whip in both House and Senate

Learning Outcomes: A Case Study



UNITED STATES
HOUSE of
REPRESENTATIVES

FIND YOUR REPRESENTATIVE
Enter your zip code: [LOOK UP](#) 

119th Congress, 2nd Session · The House is not in session.

REPRESENTATIVESLEADERSHIPCOMMITTEESLEGISLATIVE ACTIVITYTHE HOUSE EXPLAINED

Leadership

The majority party members and the minority party members meet separately to select their leaders. Third parties rarely have had enough members to elect their own leadership, and independents will generally join one of the larger party organizations to receive committee assignments. A party caucus or conference is the name given to a meeting of or organization of all party members in the House. During these meetings, party members discuss matters of concern.

[Learn more about the history of House leadership](#).

Speaker of the House



Rep. Mike Johnson
Elected by the whole of the House of Representatives, the Speaker acts as leader of the House and combines several roles: the institutional role of presiding officer and administrative head of the House, the role of leader of the majority party in the House, and the representative role of an elected member of the House. The Speaker of the House is second in line to succeed the President, after the Vice President.

Republican Leadership



Majority Leader
Rep. Steve Scalise
Represents Republicans on the House floor.



Majority Whip
Rep. Tom Emmer
Assists leadership in managing party's legislative program.



Republican Conference Chairman
Rep. Lisa McClain
Heads organization of all Republican Party members in the House.

Republican Policy Committee Chairman
(Vacant)
Heads Conference forum for policy development.

Democratic Leadership



Democratic Leader
Rep. Hakeem Jeffries
Represents Democrats on the House floor.



Democratic Whip
Rep. Katherine Clark
Assists leadership in managing party's legislative program.



Democratic Caucus Chairman
Rep. Pete Aguilar
Heads organization of all Democratic Party members in the House.



Democratic Caucus Vice Chair
Rep. Ted Lieu
Assists the Democratic Caucus Chairman.

<https://www.house.gov/leadership>

Learning Outcomes: A Case Study

Officers & Staff

+

Powers & Procedures

+

Origins & Foundations

+

Parties & Leadership

Historic Buildings & Spaces

+

Traditions & Symbols

Oral History Project

Historical Highlights

+

Senate Stories Blog

Research Tools

Related Link

[Longest Serving Party Leaders](#)

Senate Historical Office

Information provided by the [Senate Historical Office](#).

Questions about Senate History?
[Email a Senate historian.](#)

About Parties and Leadership | Majority and Minority Leaders

With each new Congress, the Democratic and Republican Conferences elect one of their members to serve as party leader. Depending on which party is in power, one party leader serves as majority leader and the other as minority leader. Both party leaders, also called floor leaders, serve as the spokesperson for their party's positions on the issues and coordinate their respective legislative strategies.

Working with the committee chairs and ranking members, the majority leader schedules business on the floor by calling bills from the calendar and keeps members of his or her caucus advised about the daily legislative program. In consultation with the minority leader, the majority leader fashions unanimous consent agreements by which the Senate may limit the amount of time for debate on a measure and divide that time between the parties.

Occupying the front-row desks on the center aisle in the Senate Chamber, the leaders pay close attention to floor action. They open and close the day's proceedings, keep legislation moving, and protect the rights and interests of party members. When several senators are seeking recognition at the same time, the presiding officer of the Senate will call on the majority leader first, then on the minority leader, and then on the managers of the bill being debated. This right of first recognition enables the majority leader to offer amendments, substitutes, and motions to reconsider before any other senator.

The position of party floor leader is not included in the Constitution. It evolved gradually in the late 19th and early 20th centuries. The position developed separately within each of the major party conferences, with the conference chairs gradually assuming the functions associated with modern day floor leaders well before creation of the title itself. By the 1910s, both parties were electing conference chairs who acted as floor leaders, and by the 1920s, these leaders were exercising the full array of responsibilities associated with modern floor leadership.

Complete List of Majority and Minority Leaders

Congress	Majority Leader	Minority Leader
51st Congress (1889–1891)	–	Arthur P. Gorman (D-MD) ¹
52nd Congress (1891–1893)	–	Arthur P. Gorman (D-MD)
53rd Congress (1893–1895)	Arthur P. Gorman (D-MD)	–
54th Congress (1895–1897)	–	Arthur P. Gorman (D-MD)
55th Congress (1897–1899)	–	Arthur P. Gorman (D-MD) ² David Turpie (D-IN)

<https://www.senate.gov/about/parties-leadership/majority-minority-leaders.htm>

Learning Outcomes: A Case Study

Home > People > Speakers of the House

RECOMMEND ✓ CITE PRINT

Speakers of the House in Numerical Order and Speaker Service

	Name	Date(s) Elected	Congress (Years)	Speaker Service in Calendar Days	State
1	Frederick Augustus Muhlenberg	Apr 1, 1789 Dec 2, 1793	1st (1789-1791) 3rd (1793-1795)	1159	Pennsylvania
2	Jonathan Trumbull	Oct 24, 1791	2nd (1791-1793)	496	Connecticut
3	Jonathan Dayton	Dec 7, 1795 May 15, 1797	4th (1795-1797) 5th (1797-1799)	1183	New Jersey
4	Theodore Sedgwick	Dec 2, 1799	6th (1799-1801)	457	Massachusetts
5	Nathaniel Macon	Dec 7, 1801 Oct 17, 1803 Dec 2, 1805	7th (1801-1803) 8th (1803-1805) 9th (1805-1807)	1913	North Carolina
6	Joseph B. Varnum	Oct 26, 1807 May 22, 1809	10th (1807-1809) 11th (1809-1811)	1225	Massachusetts
7	Henry Clay	Nov 4, 1811 May 24, 1813 ¹ Dec 4, 1815 Dec 1, 1817 Dec 6, 1819 ² Dec 1, 1823	12th (1811-1813) 13th (1813-1815) 14th (1815-1817) 15th (1817-1819) 16th (1819-1821) 18th (1823-1825)	3058	Kentucky
8	Langdon Cheves	Jan 19, 1814 ³	13th (1813-1815)	409	South Carolina
9	John W. Taylor	Nov 15, 1820 ⁴	16th (1819-1821)	563	New York

https://history.house.gov/People/Office/Speakers_Numerical_Order/

Learning Outcomes: A Case Study

What do all of these have in common?

Learning Outcomes: A Case Study

They're Tables.

Learning Outcomes: A Case Study

- So here's what I did:
 - Write a script that downloads these pages one by one
 - Extract the tabular information
 - Put it into an Excel Spreadsheet
 - Add some additional info (especially ID numbers) so that I could use the data in my research

	A	B	C	D	E	F	G	H
1	congress	year	position	chamber	name	affiliation	gentzkow_id	icpsr_id
2	43	1873	Speaker of the House	H	James G. Blaine	R-ME	43045460	760
3	43	1874	Speaker of the House	H	James G. Blaine	R-ME	43045460	760
4	44	1875	Speaker of the House	H	Michael C. Kerr	D-IN	44044610	5212
5	44	1876	Speaker of the House	H	Samuel J. Randall	D-PA	44054850	7715
6	45	1877	Speaker of the House	H	Samuel J. Randall	D-PA	45054850	7715
7	45	1878	Speaker of the House	H	Samuel J. Randall	D-PA	45054850	7715
8	46	1879	Speaker of the House	H	Samuel J. Randall	D-PA	46054850	7715
9	46	1880	Speaker of the House	H	Samuel J. Randall	D-PA	46054850	7715
10	47	1881	Speaker of the House	H	J. Warren Keifer	R-OH	47068060	5126
11	47	1882	Speaker of the House	H	J. Warren Keifer	R-OH	47068060	5126
12	48	1883	Speaker of the House	H	John G. Carlisle	D-KY	48055350	1509
13	48	1884	Speaker of the House	H	John G. Carlisle	D-KY	48055350	1509
14	49	1885	Speaker of the House	H	John G. Carlisle	D-KY	49055350	1509
15	49	1886	Speaker of the House	H	John G. Carlisle	D-KY	49055350	1509
16	50	1887	Speaker of the House	H	John G. Carlisle	D-KY	50055350	1509
17	50	1888	Speaker of the House	H	John G. Carlisle	D-KY	50055350	1509
18	51	1889	Speaker of the House	H	Thomas Brackett Reed	R-ME	51062530	7795
19	51	1890	Speaker of the House	H	Thomas Brackett Reed	R-ME	51062530	7795
20	52	1891	Speaker of the House	H	Charles F. Crisp	D-GA		2177
21	52	1892	Speaker of the House	H	Charles F. Crisp	D-GA		2177
22	53	1893	Speaker of the House	H	Charles F. Crisp	D-GA	53059180	2177
23	53	1894	Speaker of the House	H	Charles F. Crisp	D-GA	53059180	2177
24	54	1895	Speaker of the House	H	Thomas Brackett Reed	R-ME	54062530	7795
25	54	1896	Speaker of the House	H	Thomas Brackett Reed	R-ME	54062530	7795

Central Concepts

The Big Picture of Digital Strategies

Web Scraping

What it is that we're doing

Web Scraping

- Retrieving data from the web includes three steps:
 1. Scrape some websites
 2. Clean the data
 3. Save it to disk as a dataset



Simon Lee on Unsplash

Web Scraping

- Web scraping describes the process of downloading some websites that contain the data you want
- The word *scraping* is important here: you merely *scrape* the surface of the website



Simon Lee on Unsplash

Anatomy of a Website

What it is that we're scraping

Anatomy of a Website

- To understand what this means, you'll need to understand what a website is



Ilya Pavlov on Unsplash

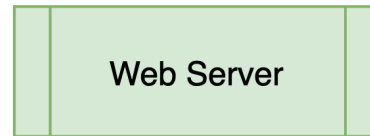
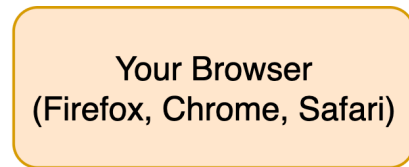
Anatomy of a Website

- To understand what this means, you'll need to understand what a website is
- What you're really after when web scraping is the **data** that sits at the heart of the website

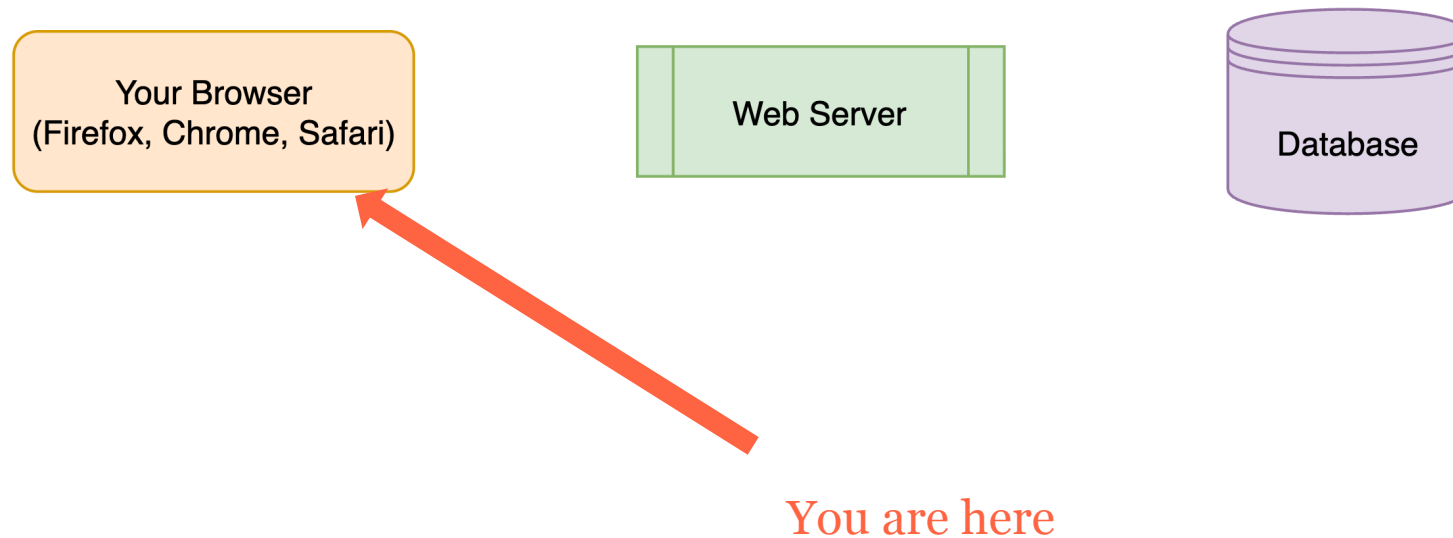


Ilya Pavlov on Unsplash

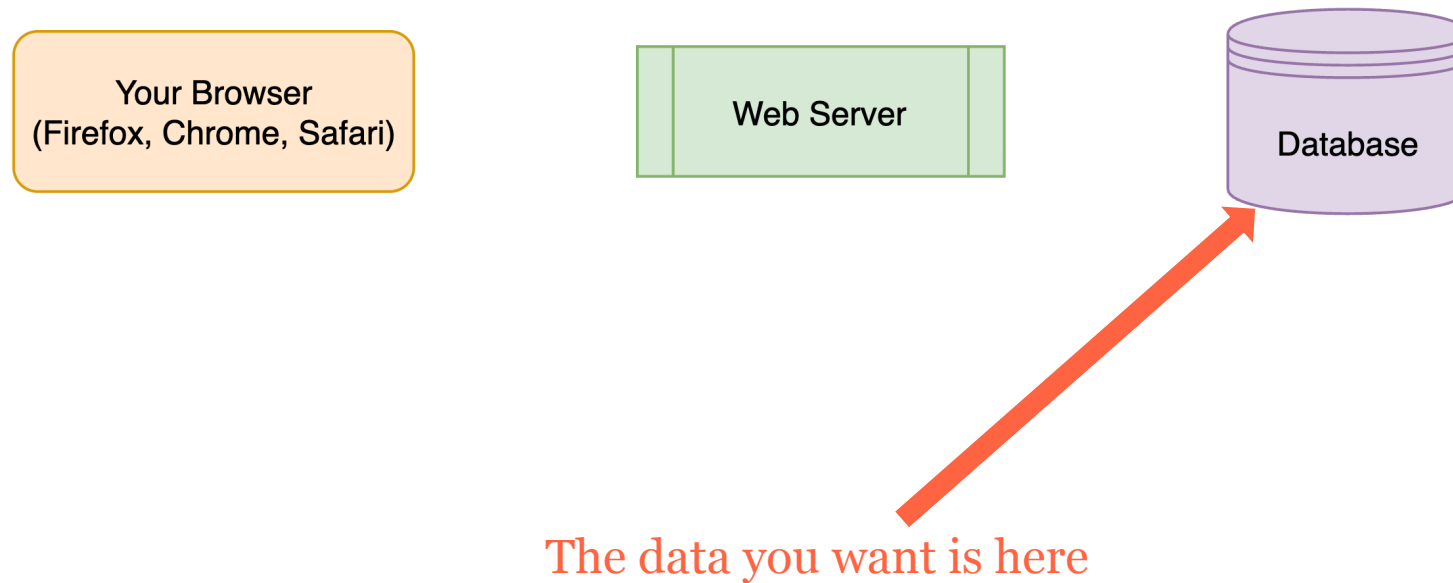
Anatomy of a Website



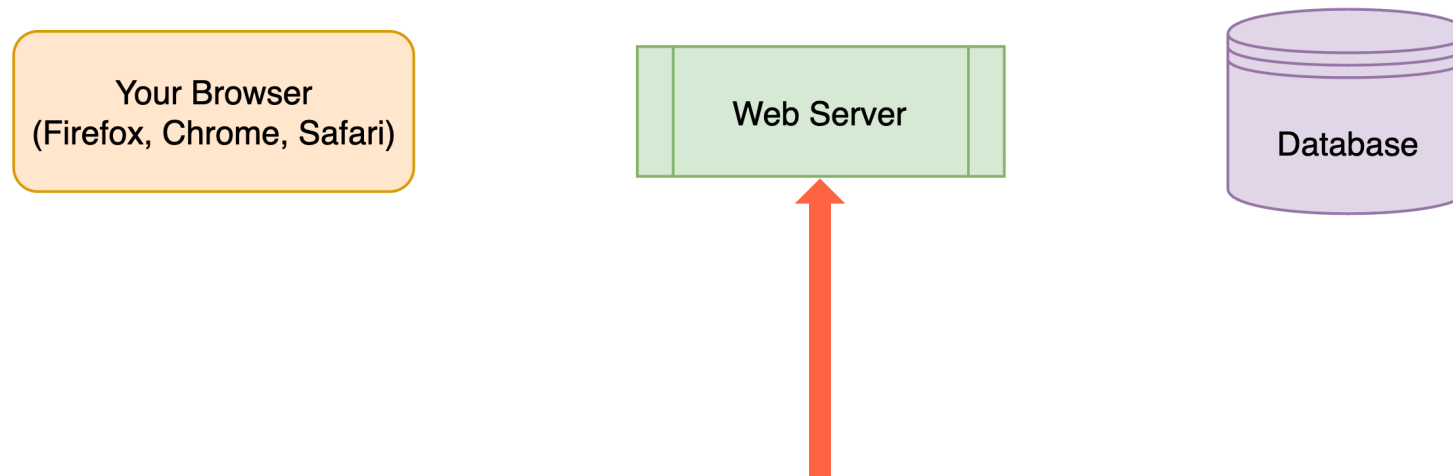
Anatomy of a Website



Anatomy of a Website

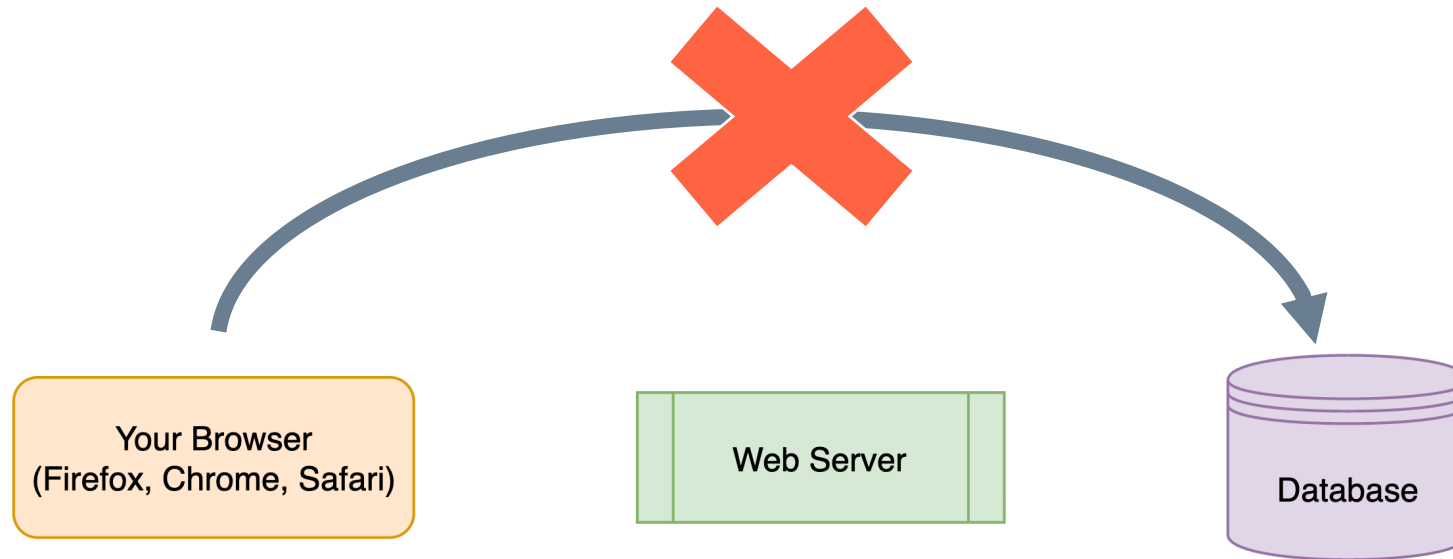


Anatomy of a Website

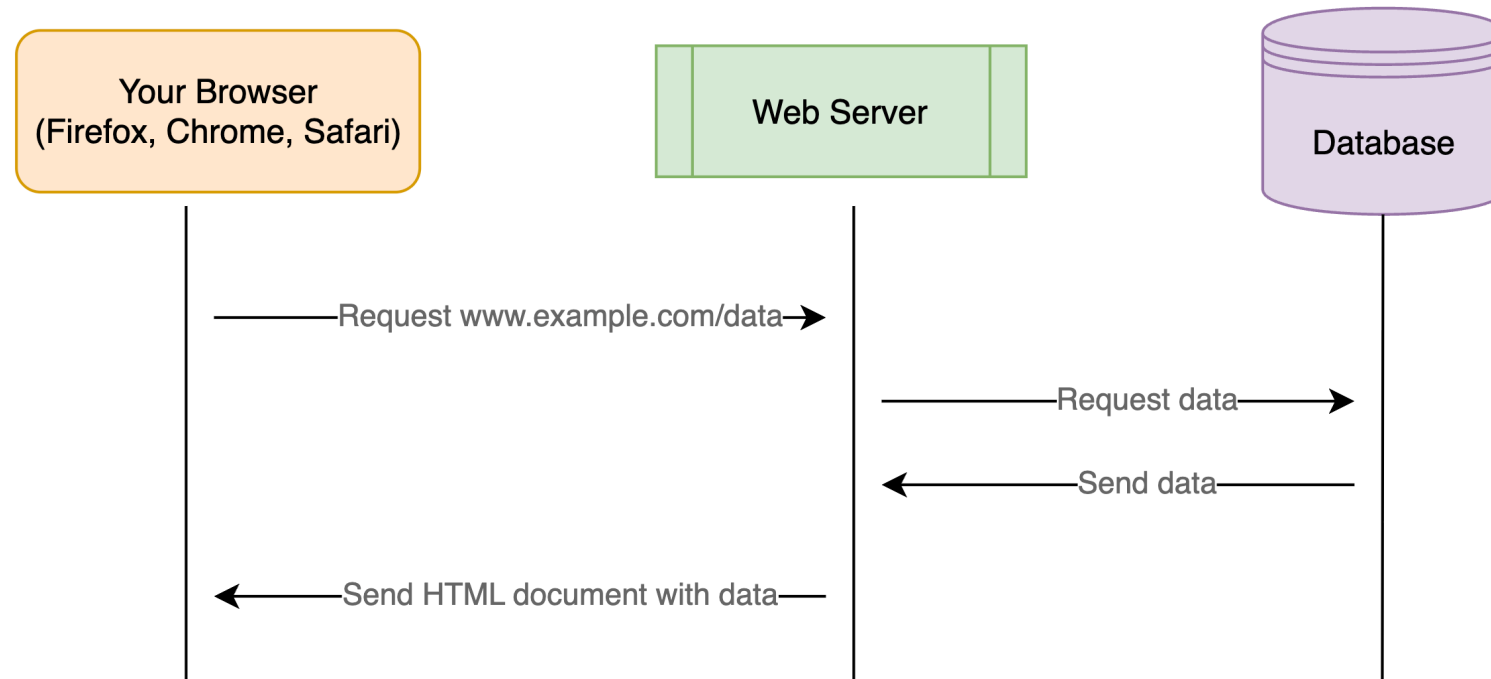


But to get it, you need to go through the server

Anatomy of a Website

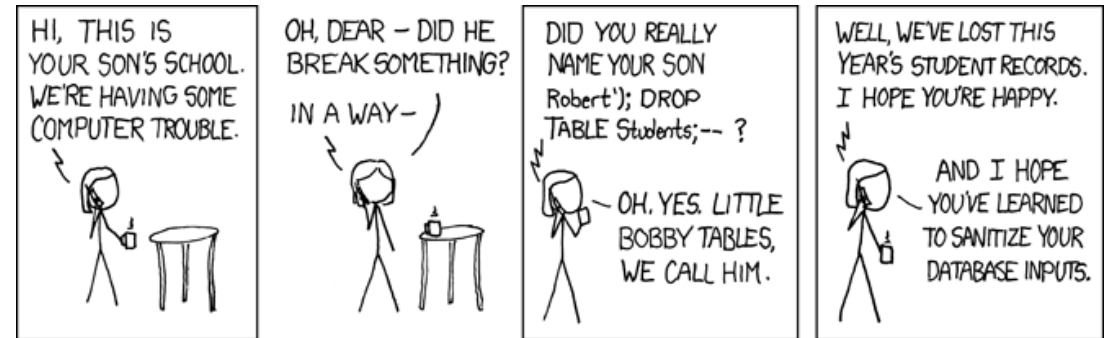


Anatomy of a Website



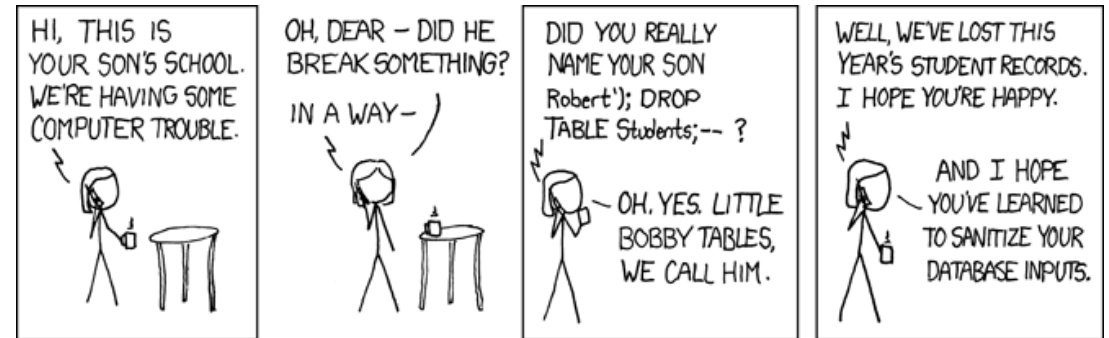
Anatomy of a Website

- Why make this so complicated?
 - Exposing a database to the internet is dangerous
 - Most databases do not only contain public information, but sensitive one, too (user accounts, passwords, etc.)
 - Therefore, you will always have to pass through a **web server**



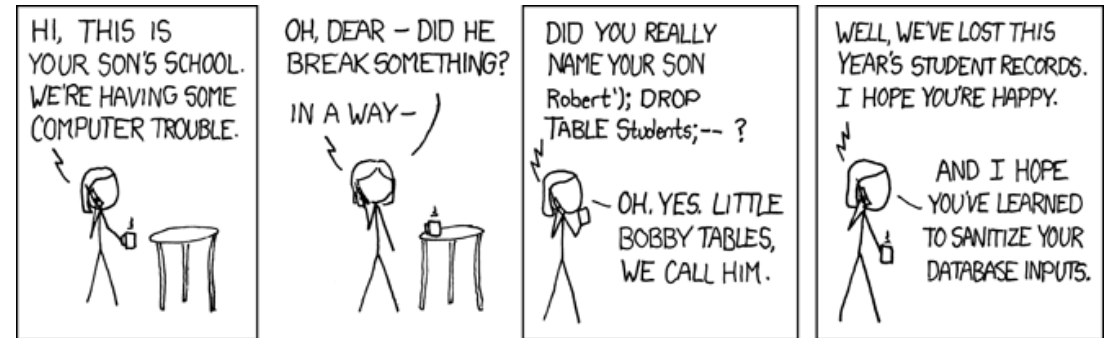
Anatomy of a Website

- Tasks of the web server
 - The web server ensures that you can access certain pages on a website (e.g., /data)
 - It also handles authentication and authorization (do you have a user account? And does your account have the rights to view this data?)
 - → It manages access to the database



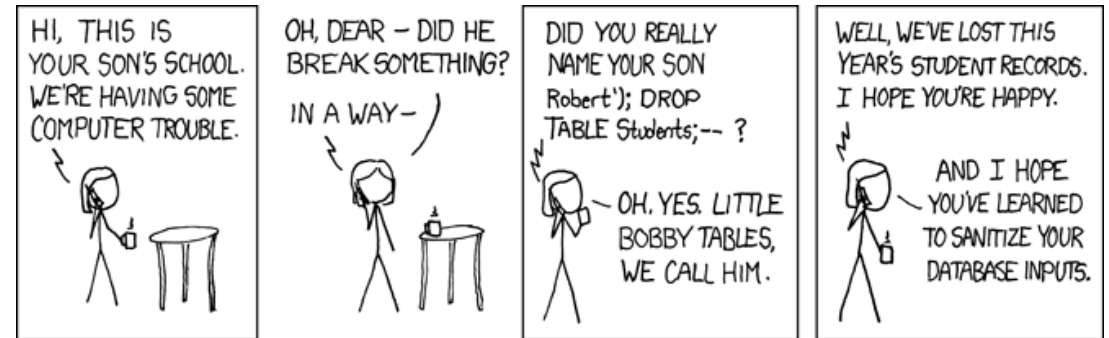
Anatomy of a Website

- A website as we understand it is **intended** to be viewed by humans. That's why there are so many additional elements (navigation, images, buttons) on it
- It is **not** intended to be automatically fetched by a computer



Anatomy of a Website

- There are exceptions, though: If a website offers you an **API (Application Programming Interface)**, you are in luck!
 - An API is intended to be read by a computer
 - It often provides the same data as the regular website, but in an already machine-readable format



HTML, CSS, and JavaScript

The building blocks of the internet

HTML / CSS / JavaScript

- The three core technologies you will interact with when scraping are HTML, CSS, and JavaScript

```
<!DOCTYPE html>
<html>
<!-- created 2010-01-01 -->
<head>
  <title>sample</title>
</head>
<body>
  <p>Voluptatem accusantium
    totam rem aperiam.</p>
</body>
</html>
```

Wikipedia

HTML

HTML / CSS / JavaScript

- HTML (HyperText Markup Language)
 - Provides **structure**
 - Defines containers for navigation, images, main page content
 - Contains the actual data, nested somewhere in between

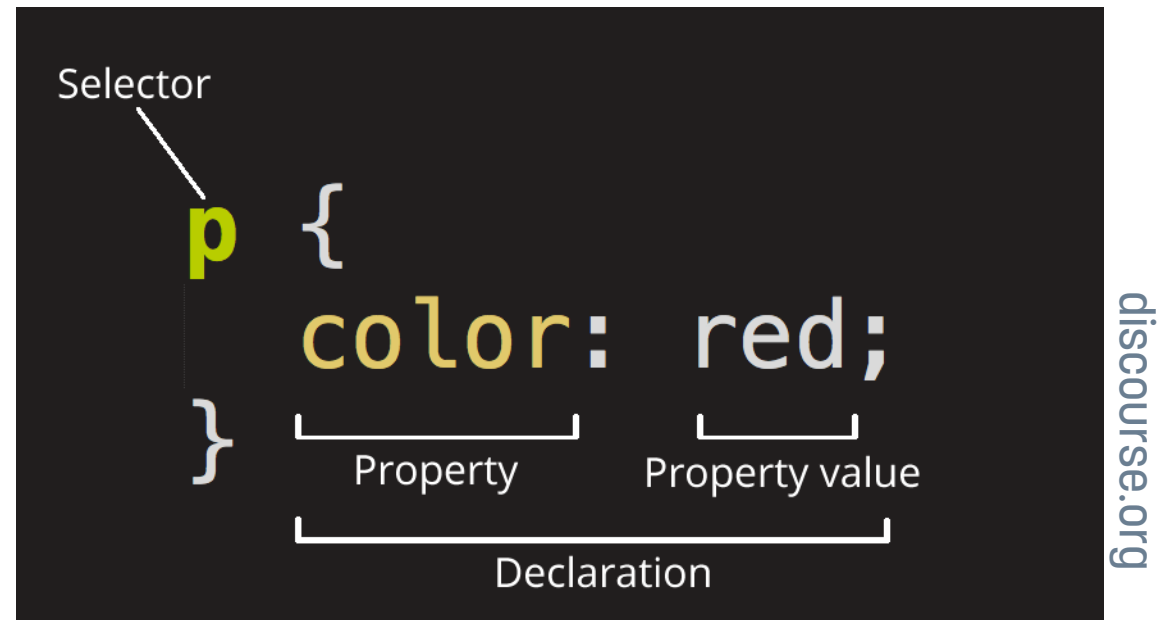
```
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</head>
<body>
  <p>Voluptatem accusantium
    totam rem aperiam.</p>
</body>
</html>
```

Wikipedia

HTML

HTML / CSS / JavaScript

- CSS (Cascading Style Sheets)
 - Provide only **styling**
 - Used to, e.g., define colors, sizes, borders, fonts, ...
 - Does not contain any of the information
 - But CSS class names can come in handy...



HTML / CSS / JavaScript

- JavaScript (not related to Java)
 - Used to make a website *dynamic*
 - For example: Load additional data after a button press without reloading the entire page
 - However, that's also a problem: web scrapers usually do not execute JavaScript → any data that is not part of the initial HTML is invisible to your scraper

```
15
16 const LOCALE = globalThis.navigator.language
17
18 const div = document.body.appendChild(document.createElement('div'))
19 const list = div.appendChild(document.createElement('ol'))
20
21 const dayNames = new Map()
22
23 for (let i = 0; i < 7; ++i) {
24   const d = Temporal.PlainDate.from({
25     year: Temporal.Now.plainDateISO().year,
26     month: 1,
27     day: i + 1,
28   })
29
30   dayNames.set(d.dayOfWeek, d.toLocaleString(LOCALE, { weekday: 'long' }))
31 }
32
33 for (const num of [...dayNames.keys()].sort((a, b) => a - b)) {
34   list.appendChild(Object.assign(
35     document.createElement('li'),
36     {textContent: dayNames.get(num)} ,
37   ))
38 }
39
```

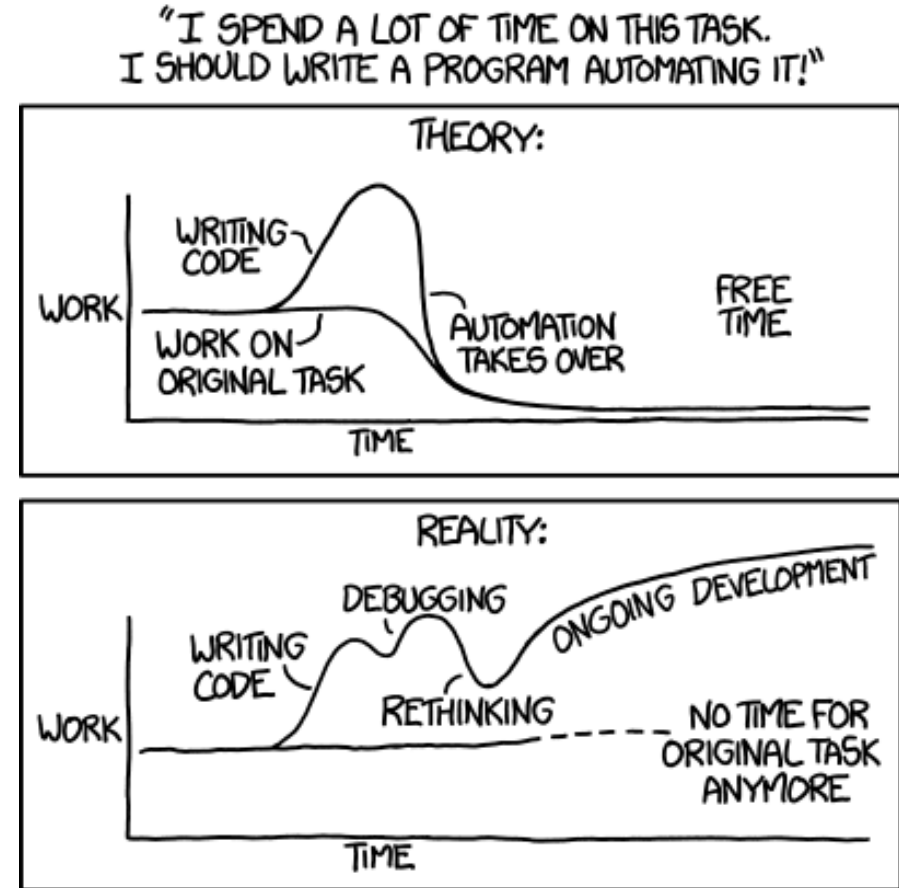
Wikipedia

Automation

How to automate the boring tasks

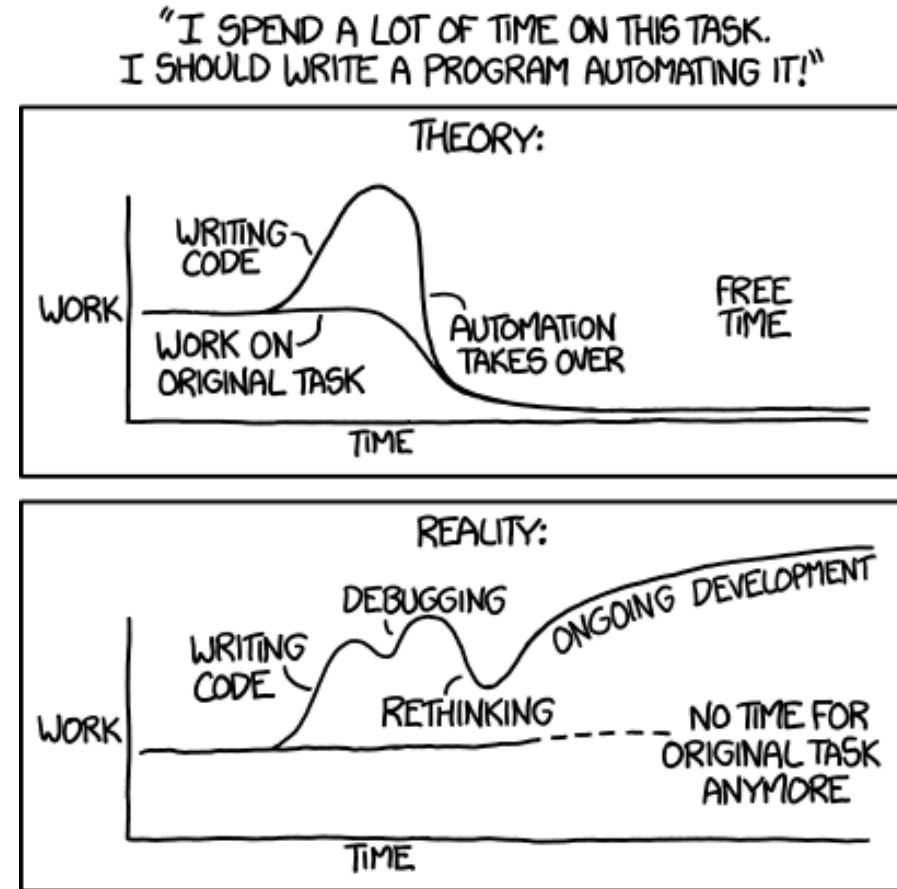
Automation

- If you have ever collected some data to satisfy your curiosity, you know how tedious it can get
- Usually, you must visit dozens of pages, extract the information, and place it into some spreadsheet
- That's where automation comes in



Automation

- Once you have told your computer how to get one website, you can tell it to do the same processes on many websites, allowing you to fetch entire datasets by letting your computer do the work
 - Pagination
 - Disparate data on multiple subpages
 - Several datasets



SPAs and Bot-Detection

The future of the internet

SPAs and Bot-Detection

- The “end boss” of web scraping are SPAs and Bot-Detection mechanisms



SPAs and Bot-Detection

- SPA (Single Page Application)
 - Very popular approach to building websites: *everything* is JavaScript
 - The original HTML only includes the reference to a big JavaScript file, your browser executes it and *that's* what will load and display the actual web page

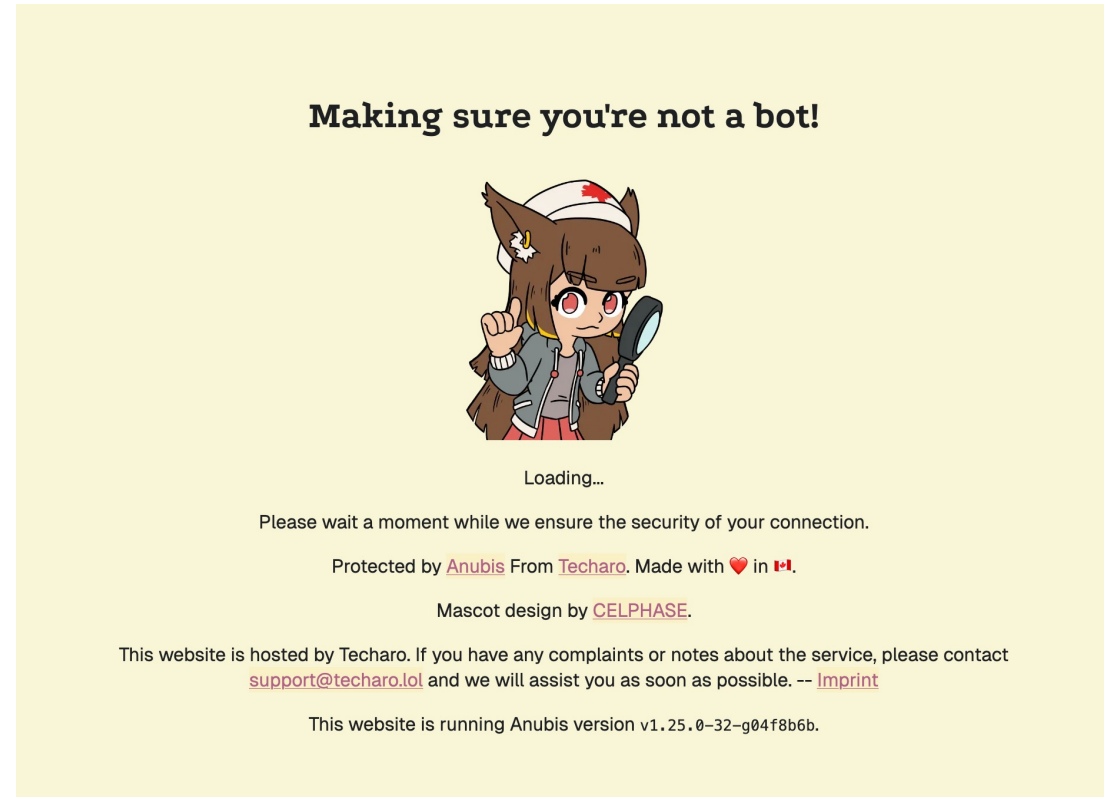
The Progressive JavaScript Framework

An approachable, performant and versatile framework for building web user interfaces.

[Why Vue](#) [Get Started →](#) [Install](#) [Get Security Updates for Vue 2 ↗](#)

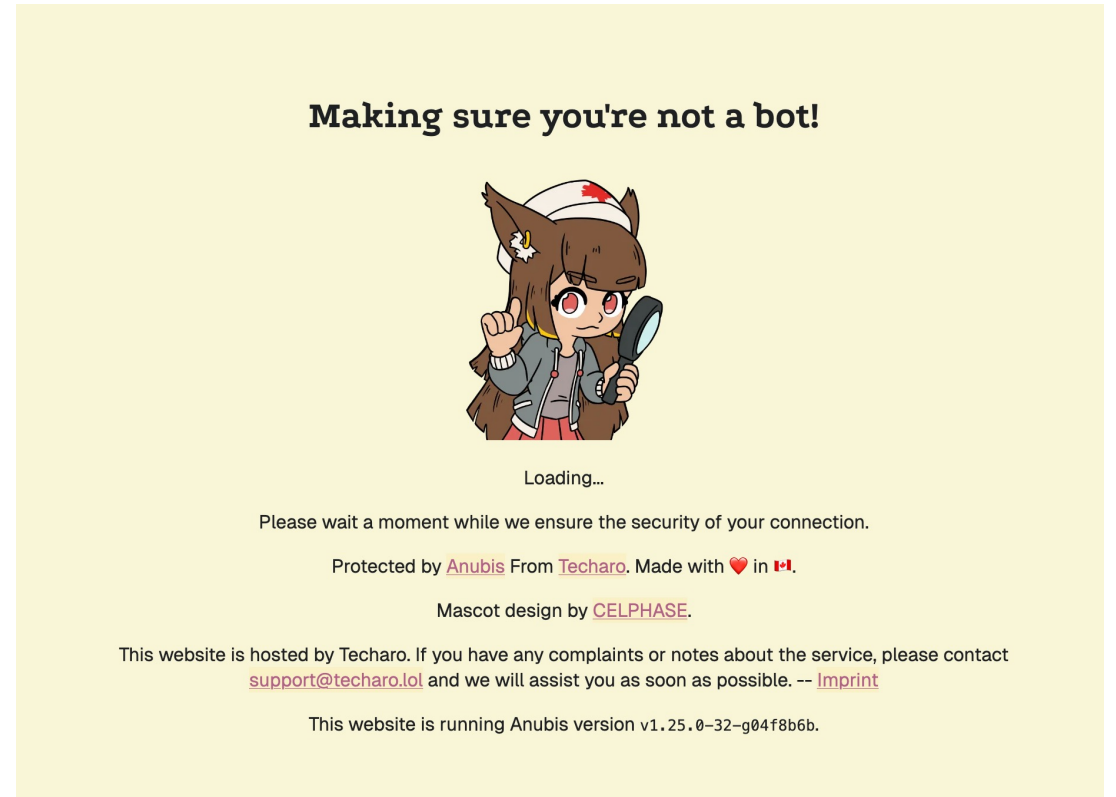
SPAs and Bot-Detection

- Bot Detection
 - Bots (that is, automated scripts, and not humans) make up **up to 50%** of all web traffic nowadays
 - The inception of chat bots has increased this a lot
 - Website owners do not like this: someone else scrapes their content, and presents it to users without naming its source



SPAs and Bot-Detection

- Bot Detection
 - This has led to a proliferation of bot-detection mechanisms
 - The three most popular:
 - CAPTCHAs
 - Cloudflare Turnstile
 - Anubis
 - Intention: Allow humans to view a website, but not automated scripts



Caveats

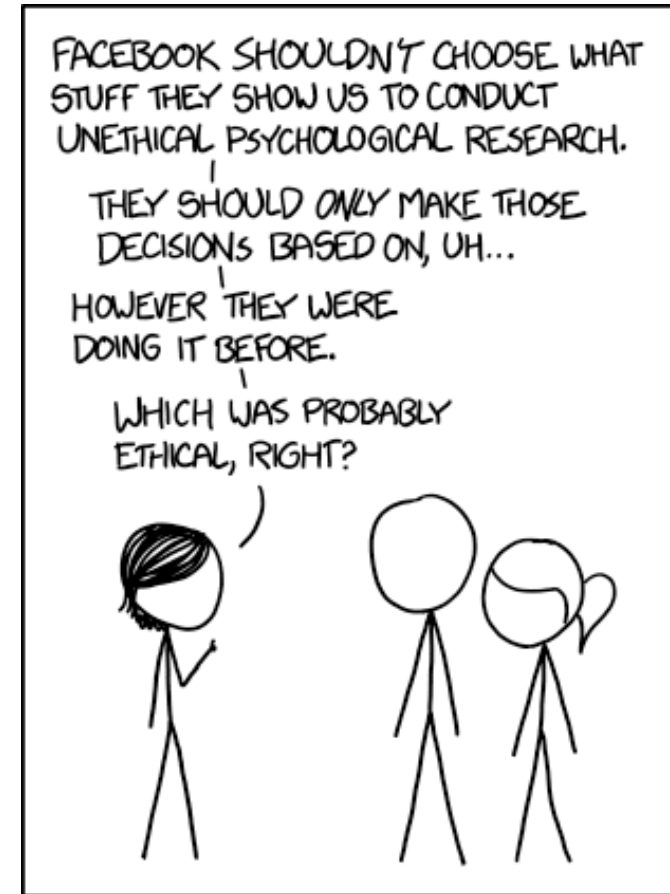
Things to keep in mind

Ethical Aspects

How not to be a d*** online

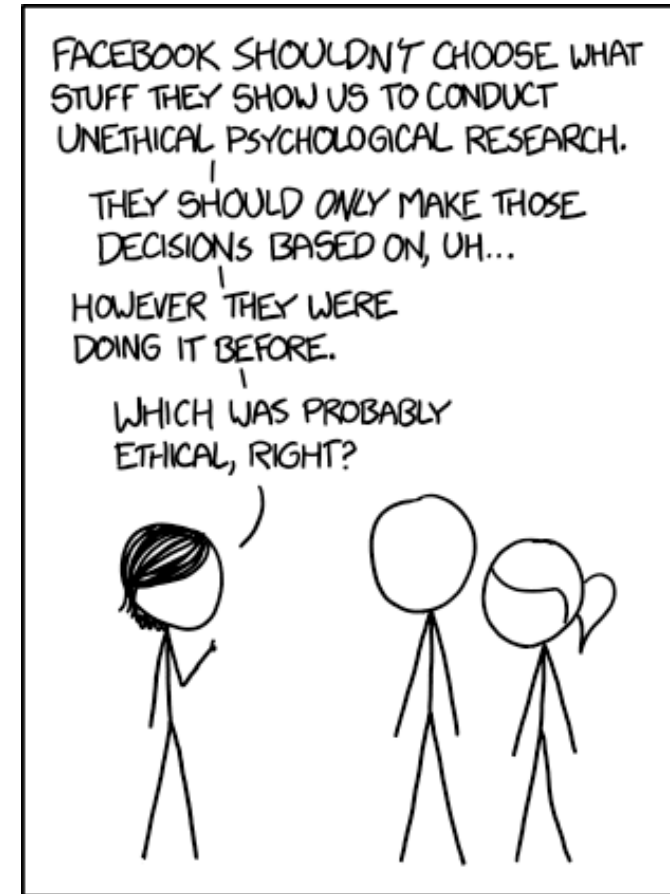
Ethical Aspects

- Just because the data is publicly available does not mean that you can just use it
- As with all other aspects of your research, you must ensure you adhere to ethical principles



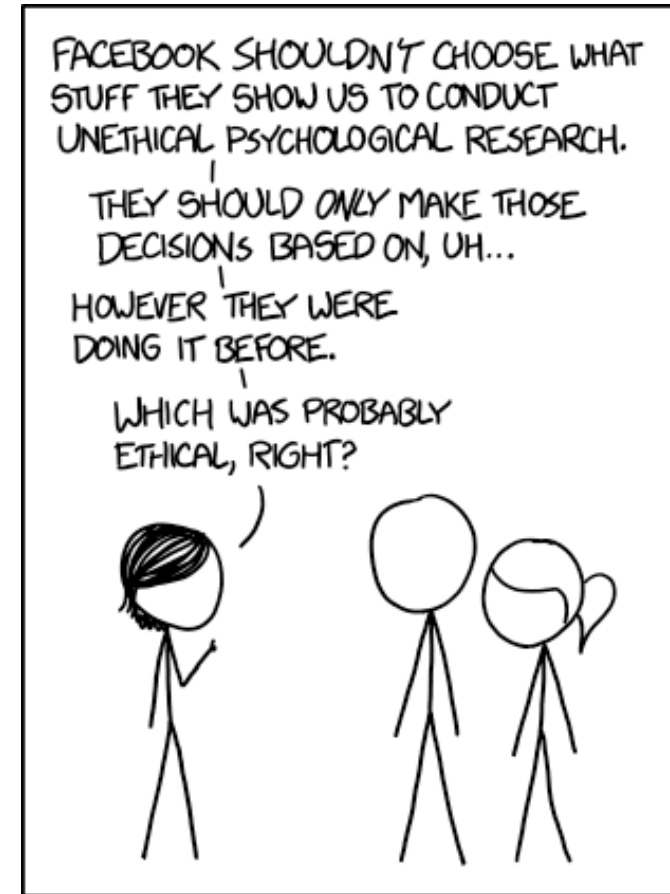
Ethical Aspects

- Belmont and Menlo reports
 - Respect for Persons
 - Beneficence
 - Justice
 - Respect for Law and Public Interest



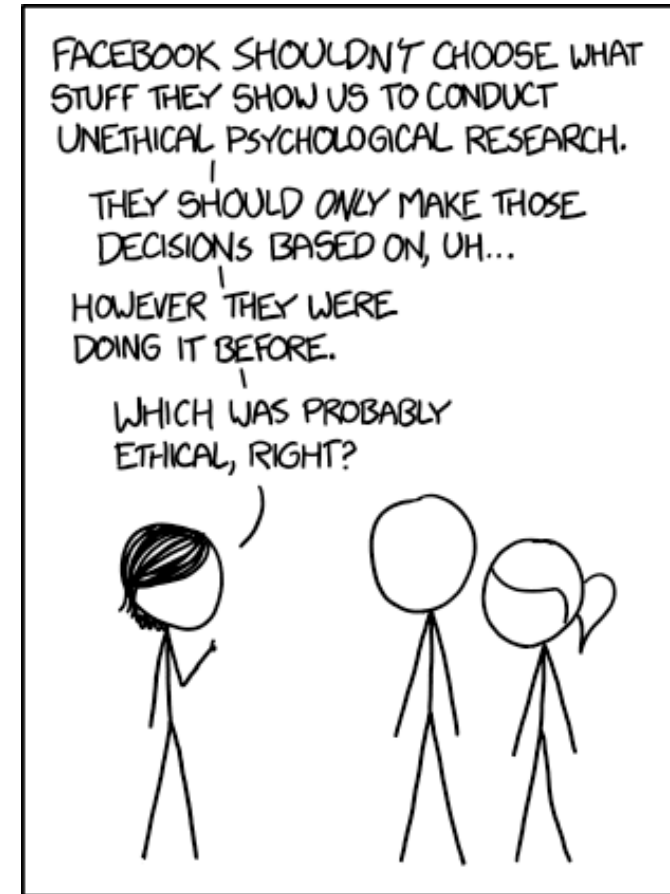
Ethical Aspects

- Respect for Persons:
 - Take the people whose information you are collecting seriously: would they be happy to learn you conducted research on them?



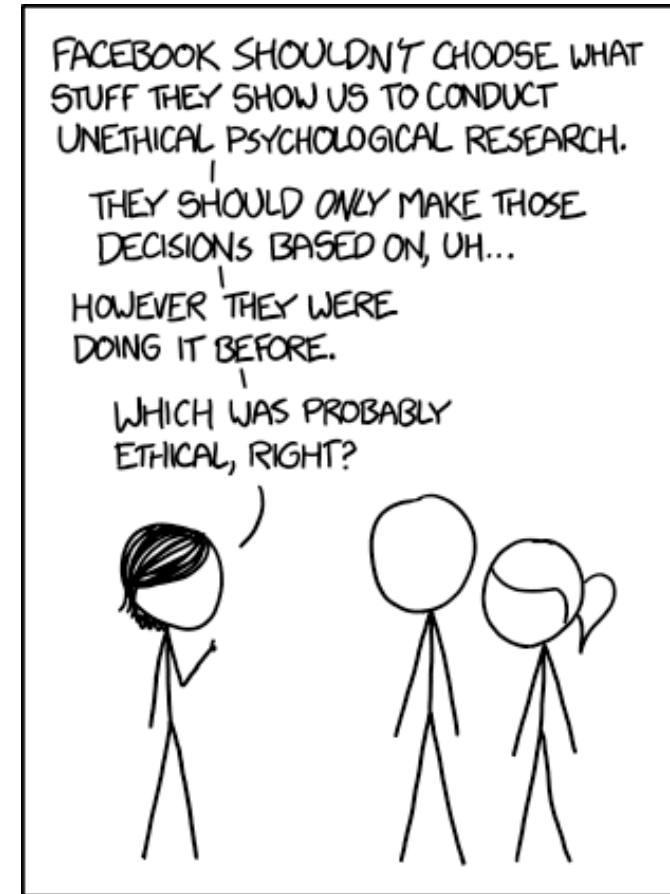
Ethical Aspects

- Respect for Persons:
 - This here is also true for the server owners themselves:
 - Do not send more requests than necessary
 - Limit the amount of requests you send to a website to not overload the web server!
 - Always be *polite*!
 - → check the robots.txt, and the ToS



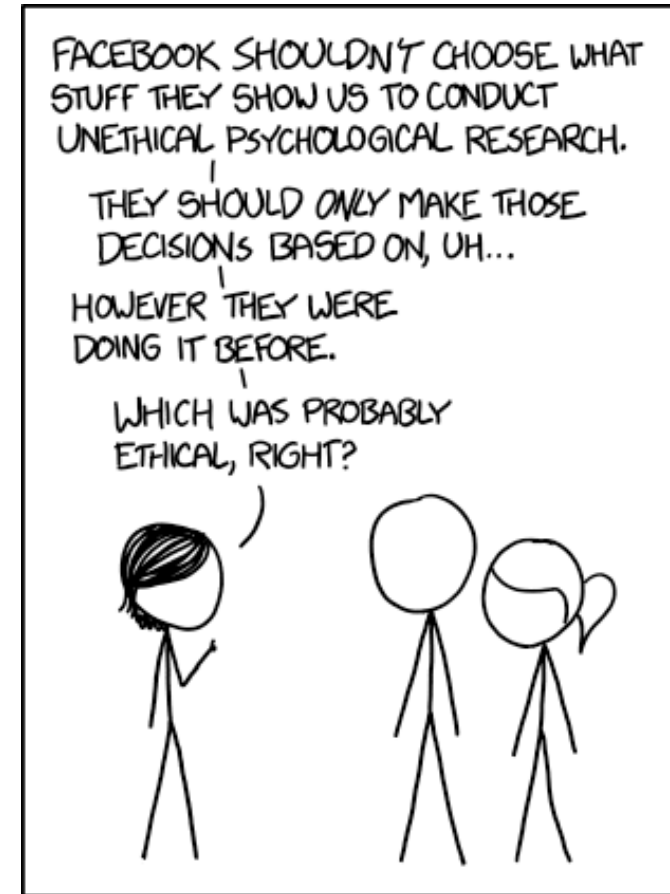
Ethical Aspects

- Beneficience:
 - Ensure that the benefits from your research will outweigh any drawbacks. Don't do research that can have detrimental impact on your research subjects' lives "for funsies"



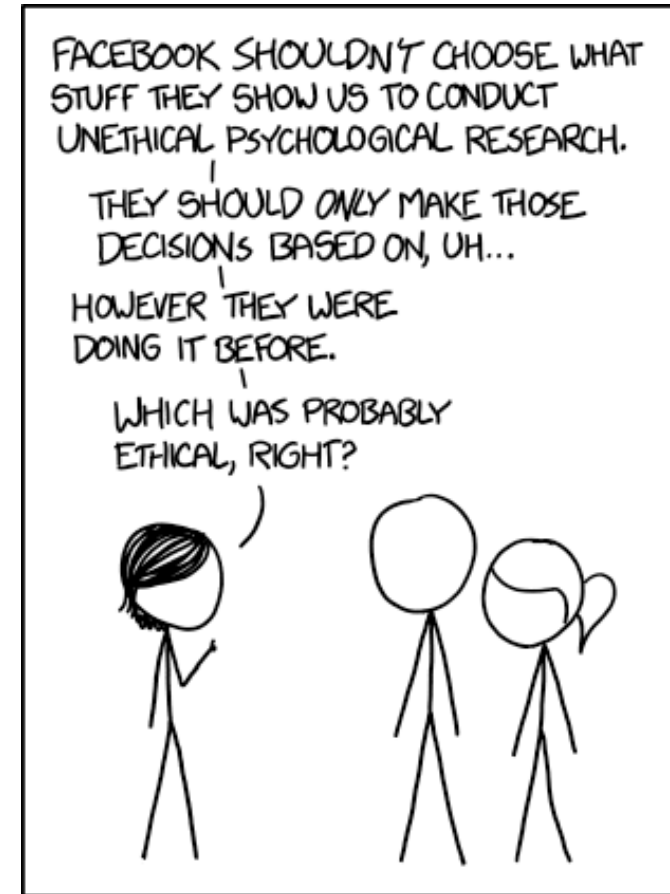
Ethical Aspects

- Justice:
 - Research can be inherently biased. Does your research benefit only a single group of people? Or does only a single group of people carry the drawbacks?



Ethical Aspects

- Respect for Law and Public Interest:
 - Does your research serve a public interest? And, does it adhere to whichever legal standards there are?

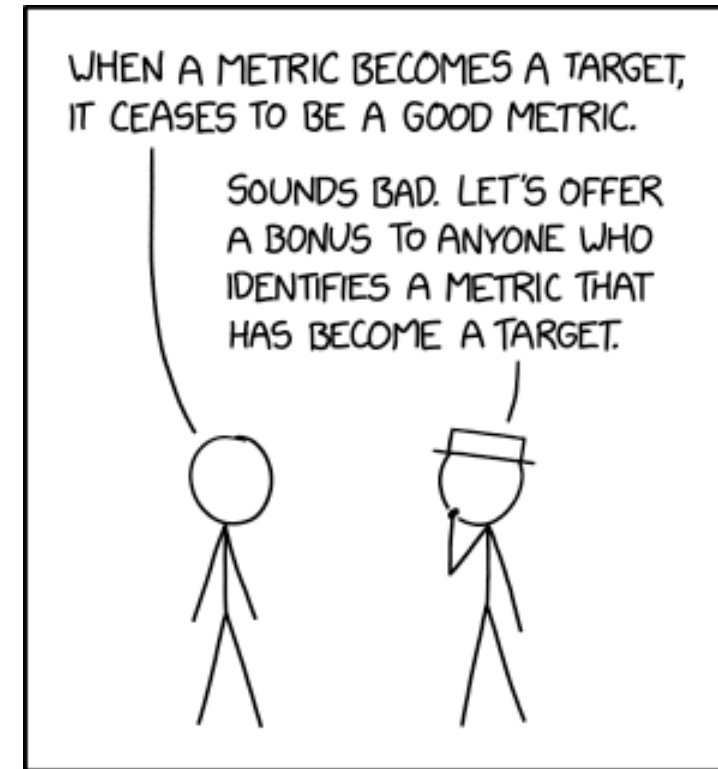


Legal Aspects

How not to tap into legal traps

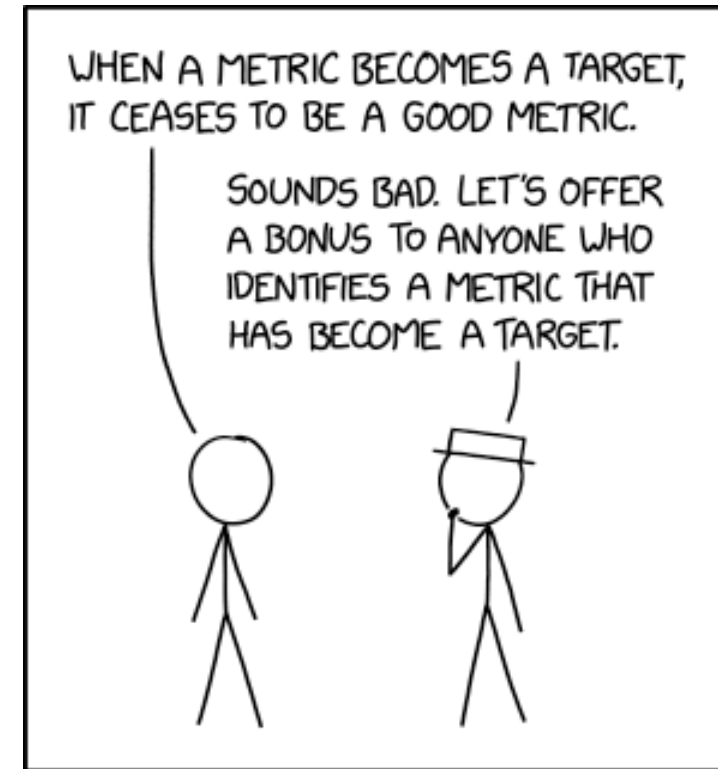
Legal Aspects

- Besides accounting for ethical aspects of your data collection, you'll need to make sure you perform *legal* research.



Legal Aspects

- Three relevant legal frameworks:
 - The website's terms of service
 - National legislation
 - International legislation



Legal Aspects

- Terms of Service:
 - Most websites nowadays have a TOS page
 - There, they outline the rules under which you can use the website
 - Typically, they include a provision **prohibiting web scraping**.

H. API Terms [↗](#)

Short version: You agree to these Terms of Service, plus this Section H, when using any of GitHub's APIs (Application Provider Interface), including use of the API through a third party product that accesses GitHub.

Abuse or excessively frequent requests to GitHub via the API may result in the temporary or permanent suspension of your Account's access to the API. GitHub, in our sole discretion, will determine abuse or excessive usage of the API. We will make a reasonable attempt to warn you via email prior to suspension.

You may not share API tokens to exceed GitHub's rate limitations.

You may not use the API to download data or Content from GitHub for spamming purposes, including for the purposes of selling GitHub users' personal information, such as to recruiters, headhunters, and job boards.

All use of the GitHub API is subject to these Terms of Service and the [GitHub Privacy Statement](#).

GitHub may offer subscription-based access to our API for those Users who require high-throughput access or access that would result in resale of GitHub's Service.

Legal Aspects

- Terms of Service:
 - Most websites nowadays have a TOS page
 - There, they outline the rules under which you can use the website
 - Typically, they include a provision **prohibiting web scraping**.

The following restrictions apply to your use of the Service. You are not allowed to:

1. access, reproduce, download, distribute, transmit, broadcast, display, sell, license, alter, modify or otherwise use any part of the Service or any Content except: (a) as specifically permitted by the Service; (b) with prior written permission from YouTube and, if applicable, the respective rights holders; or (c) as permitted by applicable law;
2. circumvent, disable, fraudulently engage, or otherwise interfere with the Service (or attempt to do any of these things), including security-related features or features that: (a) prevent or restrict the copying or other use of Content; or (b) limit the use of the Service or Content;
3. access the Service using any automated means (such as robots, botnets or scrapers) except: (a) in the case of public search engines, in accordance with YouTube's robots.txt file; (b) with YouTube's prior written permission; or (c) as permitted by applicable law;
4. collect or use any information that might identify a person (for example, harvesting usernames or faces), unless permitted by that person or allowed under section 3 above;
5. use the Service to distribute unsolicited promotional or commercial content or other unwanted or mass solicitations (spam);
6. cause or encourage any inaccurate measurements of genuine user engagement with the Service, including by paying people or providing them with incentives to increase a video's views, likes, or dislikes, or to increase a channel's subscribers, or otherwise manipulate metrics;

Legal Aspects

- General Data Protection Regulation (GDPR; or EU norm 2016/679)
 - Besides individual rights, the GDPR places hefty restrictions on what PII you can collect
 - Fortunately, there are large carve outs for research purposes

(156)The processing of personal data for archiving purposes in the public interest, scientific or historical research purposes or statistical purposes should be subject to appropriate safeguards for the rights and freedoms of the data subject pursuant to this Regulation. Those safeguards should ensure that technical and organisational measures are in place in order to ensure, in particular, the principle of data minimisation. The further processing of personal data for archiving purposes in the public interest, scientific or historical research purposes or statistical purposes is to be carried out when the controller has assessed the feasibility to fulfil those purposes by processing data which do not permit or no longer permit the identification of data subjects, provided that appropriate safeguards exist (such as, for instance, pseudonymisation of the data). Member States should provide for appropriate safeguards for the processing of personal data for archiving purposes in the public interest, scientific or historical research purposes or statistical purposes. Member States should be authorised to provide, under specific conditions and subject to appropriate safeguards for data subjects, specifications and derogations with regard to the information requirements and rights to rectification, to erasure, to be forgotten, to restriction of processing, to data portability, and to object when processing personal data for archiving purposes in the public interest, scientific or historical research purposes or statistical purposes. The conditions and safeguards in question may entail specific procedures for data subjects to exercise those rights if this is appropriate in the light of the purposes sought by the specific processing along with technical and organisational measures aimed at minimising the processing of personal data in pursuance of the proportionality and necessity principles. The processing of personal data for scientific purposes should also comply with other relevant legislation such as on clinical trials.

(157)By coupling information from registries, researchers can obtain new knowledge of great value with regard to widespread medical conditions such as cardiovascular disease, cancer and depression. On the basis of registries, research results can be enhanced, as they draw on a larger population. Within social science, research on the basis of registries enables researchers to obtain essential knowledge about the long-term correlation of a number of social conditions such as unemployment and education with other life conditions. Research results obtained through registries provide solid, high-quality knowledge which can provide the basis for the formulation and implementation of knowledge-based policy, improve the quality of life for a number of people and improve the efficiency of social services. In order to facilitate scientific research, personal data can be processed for scientific research purposes, subject to appropriate conditions and safeguards set out in Union or Member State law.

Legal Aspects

- Special case: **Transmitting data outside of the European Union**
 - As students residing in a EU member state, you are bound by European and Swedish law
 - And that explicitly **forbids** the transmission of PII outside of the European Union
 - This includes the United States
- This is especially relevant when you use U.S. services, which includes, most notably, chat bots.
- **If you even paste a name of a person into the text field of, e.g., ChatGPT, Claude, or Copilot, you are transmitting PII outside the EU, making you legally liable.**

Legal Aspects

- Special case: **Transmitting data outside of the European Union**
 - As students residing in a EU member state, you are bound by European and Swedish law
 - And that explicitly **forbids** the transmission of PII outside of the European Union
 - This includes the United States
- “But LiU uses Microsoft services!”
 - Yes, but Microsoft has a contractual obligation to keep *some* (!) parts of our data on European data centers.
 - → Always read the fine print!

Artificial Intelligence

How not to become artificially intelligent

Artificial Intelligence

- Last but not least: The use of AI for web scraping and, specifically, this course
- We explicitly **encourage** you to peruse generative AI (ChatGPT, Claude, Mistral, Copilot, etc.) for this course to achieve the learning outcomes, under the following conditions:
 - **Do** use gen AI to *explain code or concepts* to you.
 - **Do** use gen AI to discuss problems you have while solving the labs
 - **Do not** use generated code as-is. Always understand what it does, and check for errors.
 - **Do not** provide actual data to gen AI. (Contrived examples are fine.)

Quick Recap

Quick Recap

- A Case Study
- Central Concepts:
 - Web Scraping
 - Anatomy of a Website/How requests work
 - HTML, CSS, JavaScript
 - Automation
 - Single-Page Applications
 - Bot-Detection
- Caveats:
 - Ethics
 - Legality
 - AI Usage

Formalities

Structure, Projects, Examination

Course Structure

- There are ten course days in total (5 for week 1, and 5 for week 2)
- Each course day consists of a lecture and a lab
 - typically 10.15am—3.00pm (~ 2 hours each plus lunch)
 - **We want you to physically attend all lectures and labs.**
- There are three *Övningar*
 - (Friday, 24th of April, Tuesday, 28th of April, and Monday, 25th of May)
 - 3.15pm—5.00pm
 - These are **optional** and intended as extended lab sessions to help you solve especially difficult issues in your code under our supervision.

Course Structure

- Course Week 1: April 21–28, 2026
- Project 1: April 28–May 15, 2026 (17 days)
- Course Week 2: May 18–27, 2026*
- Project 2: May 27–June 12, 2026 (16 days)
- Presentations: v25 (before Midsommar)

*There may be a minor change in this week!

Examination

- The course gives 7.5 ECTS
 - Course participation grants you 1.5 ECTS pass/fail
 - The two projects grant you 6 ECTS A-F
 - Your grade is determined by the projects

Projects

- The two projects are designed to apply what you have learned directly to some real-world problem.
 - The first project shall focus on retrieving data
 - The second project shall focus on performing some initial descriptive analyses

Script Peer-Review

- In order to follow and assess your project work, we will run your R-scripts on our computer
 - Oftentimes things change in between computers
 - It is an incredibly useful skill of knowing how to write scripts that other people can run themselves
- Once you are done with your projects, we want you to perform peer-review on each other:
 - Send your final R-script to a classmate
 - They will run it to ensure that they get the same results as you
 - Then you know that we will also successfully be able to run them

Project 1

- The first project aims to demonstrate that you can write code that can extract unstructured data from a website and turn it into a dataset.
- Formal requirements:
 - 800–1,600 words (ca. 2–4p), excluding references, figures, and tables;
 - 1 working R-script (we will run this on our computers) with *extensive* (!) commentary. Describe what works, what didn't, and any extra steps.

Project 2

- The second project aims to demonstrate that you can explore a dataset using structured methods
- Formal requirements:
 - ca. 5-7 pages of text, excluding references, figures, and tables;
 - 1 working R-script (we will run this on our computers) with *extensive* (!) commentary. Describe what works, what didn't, and any extra steps.

Presentations

- In the final week of the semester, we will have project presentations
- We will talk about your two projects
- Formal setting: Carl Nordlund, you, and me, 15 minutes
- You will have both of your projects printed out, and can bring with you all notes that you want
- To get an A: Simply know what you did in your two projects. We will not ask anything that does not pertain to your individual project work.

Final Grade

- You will get all 7.5 ECTS if you participate at all lectures and labs, deliver two appropriate projects, and participate in the presentations.
- The grade for the course is the average of the two projects and the presentation:
$$\frac{Project\ 1 + Project\ 2 + Presentation}{3}$$

Office Hours

- We will both deliver office hours. (I will hold two office hours during the first week, Carl will hold two during the second week.)
- Sign up, and then be at the institute for your slot. Everyone gets 15 minutes.
- You can send us emails at any time in between.

Any Questions?

Thank you!